caugi: fast and flexible causal graph interface for R

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R package, causal inference

Motivation

- Existing workflows rely on adjacency matrices, edge lists, or general-purpose libraries like `igraph`, which are not designed for causal semantics.
- These representations are low-level, error-prone, and obscure the structure of causal graphs.
- **Goal:** provide a graph interface that is expressive, safe, and efficient for causal inference tasks.

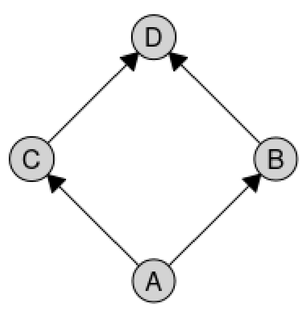
Key Contributions

- **High-performance backend:** Rust implementation with efficient graph traversal.
- **Type-safe graph classes:** Enforces structural constraints (e.g., acyclicity in DAGs) at the graph level.
- **Expressive syntax:** Define graphs using concise, composable formulas.
- **Scalable queries:** Efficient evaluation of ancestry and neighborhood relations in large graphs.

Basic Usage

Syntax like a picture in your head:

```
cg <- caugi(
  A %-->% B + C,
  B %-->% D,
  C %-->% D,
  class = "DAG"
)
plot(cg)
```



Querying and metrics

- **Relational queries**: `parents()`, `ancestors()`, `neighbors()`, etc.
- **Structural queries**: `is_acyclic()`, `is_cpdag()`, etc.
- **Adjustment and d-separation**: `adjustment_set()`, `d_separated()`, etc.
- **Graph metrics**: `shd()`, `aid()`, etc.

How it Works

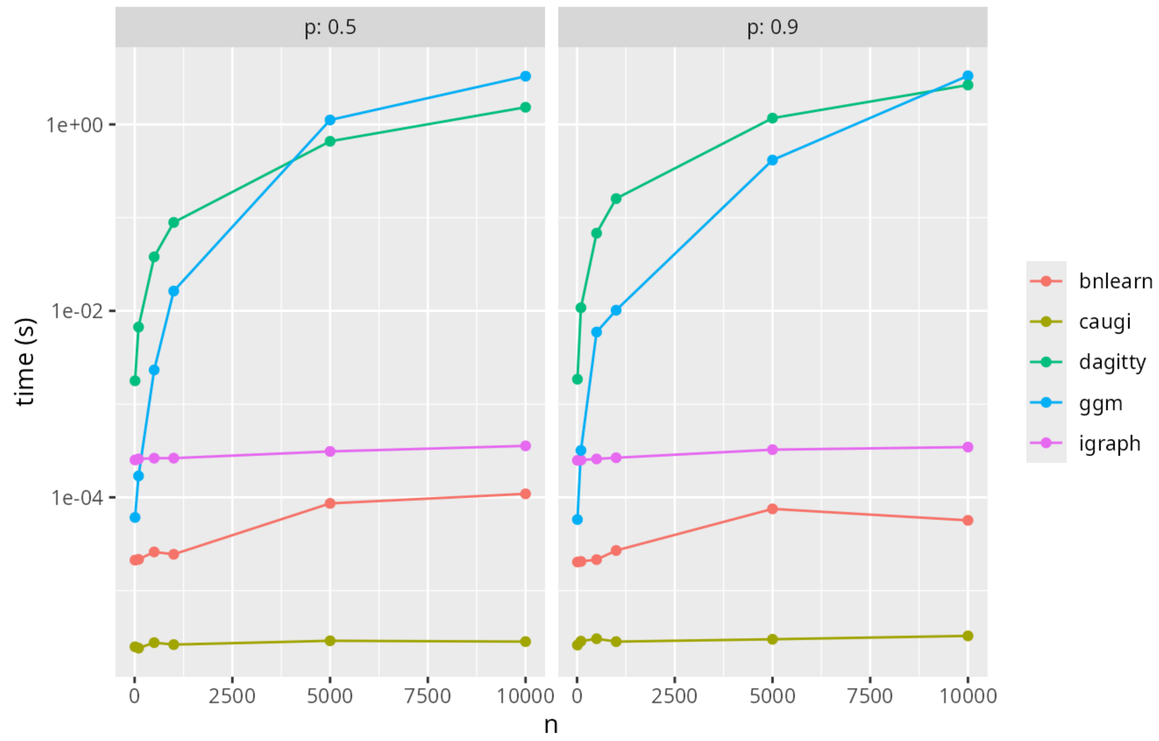
- **Backend:** Rust implementation for memory safety and performance.
- **Storage:** Compressed Sparse Row (CSR) representation
  - Memory-efficient for sparse graphs
  - Also store direct slice access giving O(1) adjacency lookup
- **Immutable design with lazy rebuild:**
  - Graph modifications trigger a rebuild with complexity O(|V| + |E|)
  - Rebuilds occur only on query (lazy evaluation)
- **Result:** Fast queries on large graphs with predictable performance and consistent state.

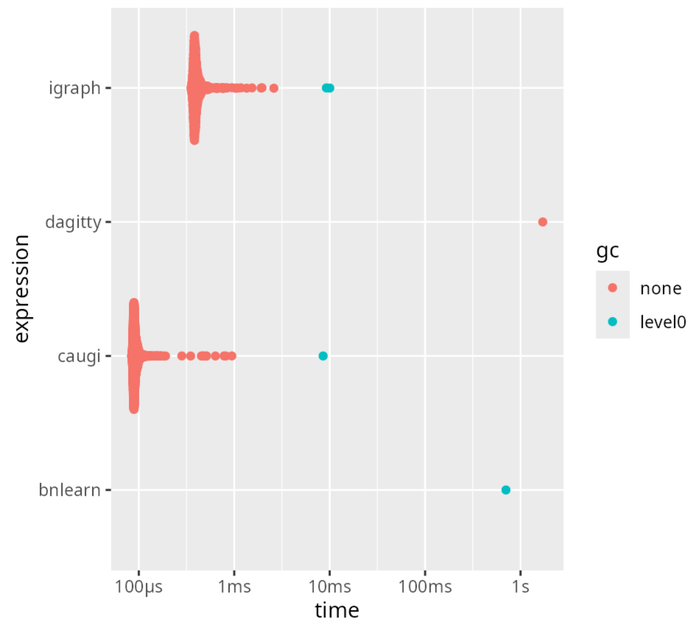
Why caugi?

- **Graph-class safe:** Think type safety, but for graphs.
  - Supports DAGs, PDAGs, ADMGs, UGs, and unknown graphs.
  - Trying to add an edge to a DAG that would create a cycle will throw an error.
- **Performance:** Fast querying of large graphs.
- **Intuitive syntax:** Express edges as simple R formulas, e.g., `A %<-> B + C`.

Benchmarks

We benchmarked `caugi` against `bnlearn`, `dagitty`, `ggm` and `igraph` for common graph queries on randomly generated DAGs of varying sizes.





Contact

- Pkgdown site: caugi.org
- GitHub: frederikfabriciusbjerre/caugi